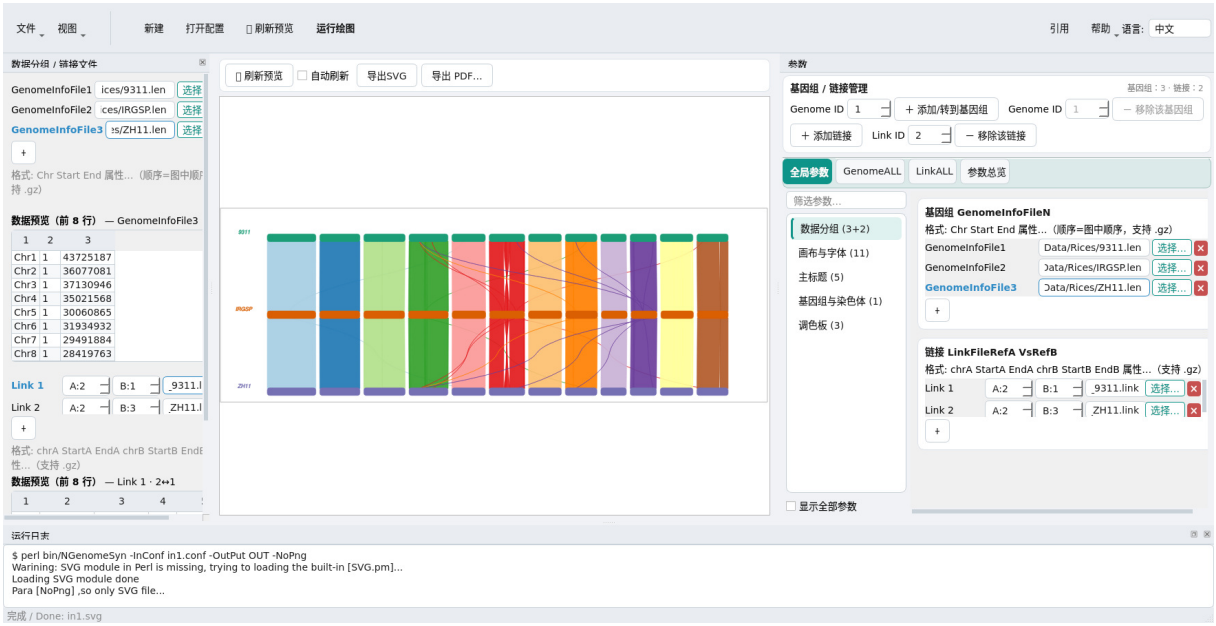
NGenomeSyn GUI Manual

GUI version: 1.50B | Engine: NGenomeSyn 1.50 (hewm2008)

1. Interface overview

The GUI is a three-column workbench: the left dock manages the two data lists (genome .len files and link .link files), the centre shows a live SVG preview (wheel zoom, drag pan, PNG/PDF export), and the right dock holds the parameter tabs (Global · GenomeALL · Genome 1..N · LinkALL · Link 1..N · Parameters summary). The run log sits at the bottom.

Every parameter shows the engine's real default, its range and bilingual descriptions (hover for details). Legacy misspelled keys are auto-corrected on load and listed in the run log.



Main window: data files + preview (left), live preview (centre), parameter tabs (right)

2. Quick start

- 1. Add at least two .len files in the left Genome files panel (row order = figure order, .gz supported).
- 2. Add .link files in the Link files panel and pick the two genome numbers with the A:/B: boxes.
- 3. (Optional) Adjust canvas, chromosome style, axis, link style and palettes in the parameter tabs.
- 4. Click Run (or F5). With Auto refresh on, every edit re-renders automatically.
- 5. Export SVG / PNG / PDF with the buttons above the preview.

3. Data formats & paths



Left dock: the two file lists; click any row to preview its first 8 rows, selection syncs across both docks

Genome info files (GenomeInfoFileN, .len)

```
Chr1  1  230000
Chr2  1  190000   fill=red stroke-width=0
```

- Whitespace separated, at least 3 columns: chr start end [attrs…]; row order = figure order.
- A row written as "end start" renders that chromosome reverse-complemented.
- Numbers must be contiguous from 1 (GenomeInfoFile1, GenomeInfoFile2, …).
- Engine rule: Start and End must not both be > 1 (the GUI pre-checks this).

Link files (LinkFileRefA VsRefB, .link)

```
Chr1  1000  50000  Chr1  900  49500
```

```
Chr1  60000  120000  Chr2  1000  61000
```

- Whitespace separated, at least 6 columns: chrA startA endA chrB startB endB [attrs…].
- The ordinal position of the line in the conf IS the Link1, Link2, … index.

Paths & .gz

- Relative paths in a conf resolve against the conf's own directory (the GUI always writes absolute paths).
- .gz inputs are pre-decompressed by the GUI (no system gzip needed) while keeping the original basename, so derived genome labels stay unchanged.

4. Special regions & zoom

- ZoomRegion=chr:start:end shows a single region; it must lie inside that chromosome.
- SpeRegionFile rows are chr start end [attrs]; SN=text adds a label, and a bare CDS/UTR/mRNA keyword in column 4 picks a preset style.

5. Parameter reference

Generated from the schema, matching the GUI; "legacy" lists misspellings that are auto-corrected.

Canvas & font

Parameter	Type	Default	Description
body	int	1200	Main drawing width (px). Total width = left+body+right.
up	int	55	Top margin (px); main title baseline depends on up/2.
down	int	25	Bottom margin (px).
left	int	100	Left margin (px); also used to centre the main title.
right	int	120	Right margin (px); base of per-genome X positions.
font-size	float	10	Base font size. Chr names ×1.2, axis labels ×1.0, main title ×1.2, genome labels ×1.4.
font-family	text	Arial	Font family passed verbatim to the SVG (e.g. Arial).
CanvasHeightRitao	float	1.0	Scales total height. Must be >0.1 or the engine silently ignores it (the 'Ritao' spelling is the engine's, keep it).
CanvasWidthRitao	float	1.0	Scales total width; also must be >0.1.
RotatePng	int	0	Rotates the exported PNG only (never the SVG).
NoPng	bool		The engine skips PNG whenever the key exists (even 0). The GUI renders PNG/PDF itself, so you normally never need it.

Main title

Parameter	Type	Default	Description
Main	text		Big centred title at the top of the figure.
MainColor / legacy: MainCor	color	blue	Title color (default blue). The 'MainCor' spelling found in old docs is a typo the GUI auto-corrects.
MainRatioFontSize	float	1.0	Title font = font-size × 1.2 × ratio.
ShiftMainX	float	0.0	Horizontal nudge for the title (px).
ShiftMainY	float	0.0	Vertical nudge for the title (px); base is up/2 - 2×font-size.

Genome & chromosomes

Parameter	Type	Default	Description
TotalGenomeNumber	int		Overrides the auto-detected genome count (advanced; normally inferred from GenomeInfoFileN / SetParaFor=GenomeN).
ZoomChr	float	1.0	>1 enlarge, <1 shrink. Must not be 0 (division by zero in the engine).
ChrWidth	int	20	Thickness of the chromosome bars (px).
LinkWidth	int	180	Vertical band between this genome and the next one (px).
ChrSpacing	int	10	Gap between neighbouring chromosomes of one genome (px).
NormalizedScale	bool	0	0=share the global scale; 1=this genome keeps its own scale.
EndCurveRadian	int	3	Rounded-cap radius of chromosome ends (px). The engine clamps values <2 back to 2.
RotateChr	float	0.0	Rotates the whole chromosome block about its start point; >0 clockwise.
ShiftX	float	0.0	Translates this genome relative to the default layout (px); additive with MoveToX.
ShiftY	float	0.0	Translates this genome relative to the default layout (px); additive with MoveToY.
MoveToX / legacy: MoveX	float		Absolute canvas X of the chromosome start (combined with ShiftX). 'MoveX' in the manual is a typo.
MoveToY / legacy: MoveY	float		Absolute canvas Y of the chromosome start (combined with ShiftY). 'MoveY' in the manual is a typo.

Genome label

Parameter	Type	Default	Description
GenomeName	text		Italic genome label on the left. Defaults to the .len basename.
GenomeNameColor	color		Defaults to this genome's chromosome bar color.
GenomeNameSizeRatio / legacy: GenomeNameRatio	float	1.0	Label font = font-size × 1.4 × ratio. 'GenomeNameRatio' in old docs is a typo.
GenomeNameShiftX	float	0.0	Horizontal nudge of the genome label (px).
GenomeNameShiftY	float	0.0	Vertical nudge of the genome label (px).

Chromosome label

Parameter	Type	Default	Description
ChrNameShow	bool		Draw the chromosome name above each bar.
ChrNameSizeRatio	float	1.0	Name font = font-size × 1.2 × ratio.
ChrNameColor	color		Defaults to this genome's chromosome bar color.
ChrNameShiftX	float	0.0	Horizontal nudge (px); additive with per-row values.
ChrNameShiftY	float	0.0	Vertical nudge (px); additive with per-row values.
ChrNameRotate	float	0.0	Extra rotation added on top of RotateChr.

Coordinate axis

Parameter	Type	Default	Description
ShowCoordinates	bool		Draw the coordinate scale for this genome.
ScaleNum	int	10	Number of axis tick divisions (N+1 ticks).

ScaleUpDown	enum		The engine only recognises "Up" (ticks above the bar); unset = above for genome 1, below otherwise.
ScaleUnit	int		Target tick spacing in bp; the derived count must land in 2..100 or the engine falls back to ScaleNum.
LabelUnit	text		Axis suffix. Auto M/K/b by default, or any custom string.
LablefontsizeRatio	float	1.0	Axis label font = ratio × font-size. Note the engine's 'Lable' spelling.
LablePrecision	int	1	Decimal places in axis labels (default 1). Real but undocumented in the manual.
RotateAxisText	float	0.0	Extra rotation of the axis text on top of RotateChr.
NoShowLabel	bool		Hides the axis numbers whenever the key exists (ticks stay). Even 0 triggers it.

Regions & zoom

Parameter	Type	Default	Description
SpeRegionFile	file		Region highlight/annotation file (.gz ok). Format: chr start end [attrs]. Rows accept SN=text or CDS/UTR/mRNA keywords.
ZoomRegion	text		Show a single chromosome region chr:start:end (must lie inside that chromosome).
SpeRegionWidthRatio	float	1.0	Bar half-height multiplier for special regions; per-row values multiply.

Links

Parameter	Type	Default	Description
StyleUpDown	enum		Case-sensitive: UpDown/DownUp/UpUp/DownDown/line. Unset = auto by genome order.
lineType	text		"line" forces straight links (equivalent to StyleUpDown=line, higher priority).
Reverse	bool	0	Swaps the start/end attachment points of the ribbon.
HeightRatio / legacy: HightRation	float	1.0	Scales the link band height; negative values are legal. The 'HightRation' typo in old docs is auto-corrected.
OutsideWidthDeta	float	-2.0	Gap between ribbon and chromosome bar (px); default -2.
ShortLinkLineRefA	float	0.0	Shortens the ribbon at the RefA end (px); negative lengthens.
ShortLinkLineRefB	float	0.0	Shortens the ribbon at the RefB end (px); negative lengthens.

Style (SVG)

Parameter	Type	Default	Description
fill	color		Genome: chromosome bar fill; link: ribbon fill (overrides the palette).
stroke	color		Genome: bar outline; link: ribbon outline.
stroke-width	float	1.0	Outline width (px), default 1; 0 = no outline.
stroke-opacity	float		0..1. Genome default 1; link default 0.92.
fill-opacity	float		0..1. Genome default 1; link default 0.92.

Color palettes

Parameter	Type	Default	Description
GenomeColorBrewer	enum	Dark2	Palette for chromosome bars (36 built-in RColorBrewer names). Unknown names crash the engine.

ChrColorBrewer	enum	Paired	Palette for link ribbons (indexed by chr row number; default Paired).
SpeRegionColorBrewer	enum	Dark2	Palette for special-region highlights (default Dark2).

6. Color palettes

The engine ships 35 built-in RColorBrewer palettes (preview them in the GUI picker). Three keys control them: GenomeColorBrewer (chromosome bars), ChrColorBrewer (link ribbons, indexed by chromosome row), SpeRegionColorBrewer (special regions). Unknown names crash the engine; the GUI validates them before running.

7. FAQ

Run fails with no-svg?

Check the engine output in the run log. Common causes: missing files, too few columns in .len/.link, or a ZoomRegion outside the chromosome — the GUI pre-check catches most of these.

The figure changed after opening an old conf — is that right?

If the conf contains legacy misspellings (HightRation, MainCor, …) the engine silently ignored them; the GUI corrects them and lists every rewrite in the log, so the figure follows the author's intent.

Does it run outside Linux/macOS?

On Windows, unzip portable Strawberry Perl into gui_runtime/perl/ (see the release notes) or install Perl normally and double-click NGenomeSynGUI.bat.

8. Citation & contact

Cite the article

If you use NGenomeSyn in your work, please cite:

```
He W, Yang J, Jing Y, Xu L, Yu K, Fang X. NGenomeSyn: an easy-to-use and flexible tool
https://doi.org/10.1093/bioinformatics/btad121
```

See the Citation section of the project README for the full article details.

Contact

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https://github.com/hewm2008/NGenomeSyn
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